Department of Computer Science

CS5500 MSc Dissertation
Coursework for 2023/24

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Assessment Title	Master's Dissertation
Module Leader	Dr Sarath Dantu
Distribution Date	26th Jan 2024
Submission Deadline	11th Sep 2024, 11:00
Feedback by	NA
Contribution to overall module assessment	100%
Indicative student time working on assessment	600 hours
Word or Page Limit (if applicable)	12,000 words (excluding title page, abstract, acknowledgements, table of contents, references, and appendices)
Assessment Type (individual or group)	Individual

MAIN OBJECTIVE OF THE ASSESSMENT

The aim of this assessment is to demonstrate advanced knowledge and critical skills, in an agreed topic area relevant to your master's programme.

DESCRIPTION OF THE ASSESSMENT

Through your written dissertation (submitted as a single PDF file) you will be demonstrating that:
- you can critically identify a research problem in a topic area related to your master's programme.
- you can critically identify and apply appropriate methods for the chosen problem.
- you can critically describe, evaluate, and present the results for the chosen problem.
- you can clearly communicate your work to your reader.
- you have engaged with the University's ethics process and where required have the proper ethical approval for your study.

For further details on how each criterion is assessed please see below.

LEARNING OUTCOMES AND MARKING CRITERIA

Your MSc dissertation (Task 2) will be assessed using the criteria listed below.

Learning Outcomes	Marking Criteria
LO1: Critically evaluates relevant research and scholarship within the field of study	F (Fail): Fails to provide any relevant literature review.
	D band (narrow fail): Insufficient evidence of relevant literature review for the field of study.
	C band: Provides minimal but sufficient background to understand the field of study, demonstrating the ability to critically evaluate the essential contributions in the field.



	B band: Literature review is thorough providing a critical analysis of the research area and is used to derive clear objectives relevant to the problem.
	A band: Critical overview of the research area is presented, explaining the gaps and the need for current research aims and objectives demonstrating an excellent grasp of the subject
LO2: Chooses and applies appropriate techniques and enquiry, given the research problem (or question) addressed	F (Fail): Fails to identify relevant techniques/methods suited for the problem.
	D band (narrow fail): Fails to apply relevant techniques/methods for the chosen problem.
	C band: Applies appropriate research methods, providing a basic description of the methods.
	B band: In addition to the previous criteria, a logical research plan is proposed with a very good discussion of each method and it's suitability for the research questions.
	A band: In addition to the previous criteria, the choices of methods is critically discussed and their relevance to the objectives/research questions is critically justified with the pros and cons of each method established.
LO3: Proposes, presents, and critically evaluates a solution to the research problem addressed	F (Fail): Fails to present and explain any relevant results.
	D band (narrow fail): Results relevant to the problem are present but the presentation is very weak.
	C band: Critically presents results/solutions relevant to the proposed objectives with a basic description of the proposed solutions/results.
	B band: In addition to the previous criteria, there is a logical presentation of the findings with good use of visual representations (graphs/tables) for quantitative research and cogent arguments for quantitative research.
	A band: In addition to the previous criteria, the proposed solutions/results are critically evaluated, drawing on relevant literature where required, explaining the success/failure of findings in addressing the research objectives and their significance for the proposed research problem.
LO4: Clearly communicates the above achievements to the reader	F (Fail): The dissertation has major language and formatting issues with very poor structure and presentation.
	D band (narrow fail): The dissertation is very difficult to comprehend with a lot of language issues.
	C band: While the idea of the project is understandable, the structure and presentation of the dissertation could be improved. Referencing is appropriate, with minor language issues.
	B band: Dissertation is written clearly, all sections are easy to understand, consistent referencing style, demonstrating the ability to develop and communicate a master's level project.
	A band: In addition to the above criteria, the dissertation is exceptionally well written, communicating complex ideas clearly and concisely to expert and non-expert audience with excellent use of visual representations, creating a logical flow through the sections, with thorough and consistent referencing with only very few grammatical/typographical issues.



LO5: Demonstrate engagement with the University's ethics process and, where required, gain formal ethical approval from the University and evidence this in the dissertation report	Fail: No engagement with the ethics process.
	Pass: Engages with the ethics process; obtains either an ethical approval letter (included in the dissertation or attached as an additional file) or a clear statement is present in the dissertation explaining that the project does not need ethical approval.

We are aware that a small minority of students are purchasing dissertations rather than writing themselves. So-called contract cheating is considered an act of academic misconduct (SR6:21). If your supervisor suspects you of engaging in contract cheating (or any other type of academic misconduct), as per Senate regulation 4 (17), they reserve the right to require you to attend a *viva* to demonstrate that the work is your own.

Format of the Assessment

You will prepare and submit the **MSc Dissertation (Task 2)**, using the template provided on Brightspace. Your dissertation should not exceed 12,000 words (±10%) (excluding title page, abstract, acknowledgements, table of contents, references, and appendices).

You are encouraged to use the Harvard referencing style (example in Task 1 template), but the use of any other citation style is not a problem as long as you use a consistent style throughout the document, using a reference management software like Zotero/Mendeley/RefWorks or any other software that is convenient for you and is available for free. Lecture 2 in Term 3 (week 37, date to be confirmed), will give you a demonstration on how to use Mendeley reference manager.

You must write your coursework using Microsoft Word (or similar software packages e.g. Pages, LibreOffice Writer, LaTeX, etc.) and submit it as an Adobe PDF file on WISEflow by **11th of September 2024, 11:00.**

Any code developed during your dissertation should be included as text (not as images) in the Appendix.

Submission Instructions

You must submit your coursework as a PDF file on WISEflow on **11th of September 2024, before 11:00**. You can follow the link to WISEflow through the module's section on Brightspace or login in directly at https://uk.wiseflow.net/brunel. The name of your file should follow the normal convention and must therefore include your student ID number (e.g., 0612345.pdf). It can also include the module code (e.g., CS5500_0612345.pdf).

You will be submitting the following files:

File	Description	File type	Optional/Mandatory
Main Dissertation ("StudentID.pdf")	Your report should be contained in one document, including all the figures, tables, and any code, as a PDF document.	pdf	Mandatory
Ethical approval letter	Either as part of the main document or additional file	pdf	Mandatory
Additional files: Software/code compressed zip folder	If you have developed a software or analysis pipeline using any of the popular programming languages (R/python, etc.) using jupyter-lab or Rmarkdown notebooks, you can upload these files and the sample data files (subsample the data so that the data size does not exceed 5GB) as a compressed appendix file.	Zip folder	Optional



If there are any problems with the submission, please visit the Taught Programmes Office in room 056 of the Michael Sterling building.

We STRONGLY suggest that you attempt to submit your dissertation a day early. It will be a large document and sometimes will take several attempts to upload.

ALL SUBMITTED DISSERTATIONS WILL BE CHECKED FOR PLAGIARISM!

Avoiding Academic Misconduct

Before working on and then submitting your coursework, please ensure that you understand the meaning of plagiarism, collusion, and cheating (including contract cheating) and the seriousness of these offences. Academic misconduct is serious and being found guilty of it results in penalties that can reduce the class of your degree and may lead to you being expelled from the University. Information on what constitutes academic misconduct and the potential consequences for students can be found in Senate Regulation 6.

You may also find it useful to read this PowerPoint presentation which explains, in plain English, the different kinds of misconduct, how to avoid (even accidentally) committing them, how we detect misconduct, and the common reasons that students give for engaging in such activities.

If you are experiencing difficulties with any part of your studies, remember there is always help available:

- Speak to your personal tutor. If you're not sure who your tutor is, please ask the Taught Programmes Office (TPOcomputerscience@brunel.ac.uk).
- Alternatively, if you prefer to speak to someone outside of the Department you can contact the Student Support and Welfare team.

Expectations of Artificial Intelligence Use

Generative AI tools such as ChatGPT, Bard, GitHub Copilot, Alpha Code etc., are powerful tools and might help you in building your understanding of the research topic. But please beware that these technologies are not error free, biased, may be inaccurate, contain fabricated information and may mislead you with fake references (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10277170/). Therefore, please use generative AI tools cautiously, apply critical thinking, check multiple literature sources, understand the generated code, develop your understanding of the topic/code, and implement your own text/code. Any use of AI-tools for example proof-reading, should be acknowledged (e.g.: ChatGPT for proofreading/algorithm design). **Do not present any AI-generated content as your own and any such practice can lead to academic misconduct.** The University has general guidance on using artificial intelligence in your studies.

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Late Coursework

The clear expectation is that you will submit your coursework by the submission deadline stated in the study guide. In line with the University's policy on the late submission of coursework (revised in July 2016), coursework submitted up to 48 hours late will be accepted but capped at a threshold pass (D- for undergraduate or C- for postgraduate). Work submitted over 48 hours after the stated deadline will automatically be given a fail grade (F).

Please refer to the Computer Science student information pages and the Coursework Submission Procedure pages for information on submitting late work, penalties applied and procedures in the case of Extenuating circumstances.

